

Are You Tired? I am: Trying to Understand Privacy Fatigue of Social Media Users

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ABSTRACT

Privacy fatigue is a new perspective to explain the discrepancy between individual attitude and behavior, *i.e.*, privacy paradox. Fatigued users are powerless about privacy decisions, and thus "forced to accept" or "regardless of it". In order to understand social media users' privacy fatigue comprehensively, we use China's mainstream social media Weibo as the background to conduct experiments on 428 users. Specifically, we propose and verify the perceived antecedents and behavioral consequences of the privacy fatigue model based on the Stimulus Organism Response (S-O-R) theory, taking personality traits as the moderating factor. Based on collectivist culture, the study provides complementary evidence to existing research on cultural factors. We hope it will enrich the privacy fatigue theory, helping researchers and service providers pay attention to users' potential fatigue and implement usable privacy design in real systems to alleviate privacy fatigue.

CCS CONCEPTS

• **Human-centered computing** → *HCI theory, concepts and models; Computer supported cooperative work; Social media.*

KEYWORDS

Privacy Fatigue; Stimulus Organism Response (S-O-R) Theory; Big Five Personality Traits

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1 INTRODUCTION

Studying the attitudes and behaviors of social media users towards privacy has been an important area of research [22]. Especially since the COVID-19 pandemic, people have increasingly relied on the Internet to socialize. "The geese leave a mark", along with the user's operation, personal privacy information is still in cyberspace, and it is easy to collect and spread. A recent global study by IBM

Security found that the pandemic resulted in a more expensive data breach cost of \$4.96 million [1].

At the same time, the measures required to protect individual privacy have become increasingly complex, such as password updates, vague privacy policy terms, and privacy settings that are difficult to find and understand [24, 35]. Some users no longer consider taking measures to protect their online privacy because they think the measures are doomed to failure [72]. Users' negative emotions and negative use patterns have become the new normal of society in the era of big data. Privacy fatigue is a negative emotion, which reflects that users are tired of personal privacy decisions and are unwilling to actively manage and control personal privacy information, which attempts to provide a new analytical perspective for privacy paradox from the internal transmission mechanism between privacy attitude and privacy behavior [15].

However, there are still few theoretical studies on privacy fatigue [16, 24]. While these studies provide valuable insights into the importance of identifying privacy fatigue, the characteristics that may be contributing to social media privacy fatigue remain unclear. We try to supplement and further broaden the existing work. Theoretically, it is necessary to fully understand and verify the privacy fatigue of social media users, including perceived antecedents and behavioral consequences caused by fatigue. On the other hand, since personality traits are the background component of rational behavior theory [23], we explore the relationship between users' personalized factors and privacy fatigue. Above all, the main contributions of our work are as follows:

(1) Instead of exploring the benefits of social media for users, we consider the dark side of social media, namely the potential information security risks. We will abandon the traditional privacy computing theory based on the hypothesis of Rational Economic Man [4], and start from its opposite, based on the privacy fatigue theory. Considering both objective environmental factors and subjective personal factors, we explore the occurrence mechanism and interaction relationship of users' privacy fatigue behavior from the cognitive and behavioral levels.

(2) Fatigue is a highly individualized measure [62]. We will consider privacy fatigue from the perspective of personality traits, exploring the factors and moderation influence to better understand the generation mechanism of social media users' privacy fatigue.

(3) Privacy decisions are closely related to cultural characteristics, and China's social media platforms and users have their own characteristics [27], we will choose China's mainstream social media Weibo platform as the research background. We study the privacy fatigue theoretical model from a larger context perspective. The

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results provide new evidence to supplement the previous discussion on the factors of collectivist culture and individualism culture.

2 RELATED WORK

Research has introduced various concepts related to security fatigue in cyberspace [24, 38, 54]. All these concepts represent psychological and decision-making fatigue for online privacy and security [58]. Tanimoto et al. [61] discussed the conceptual modeling technology of the severity of privacy fatigue and proposed a model to visualize the degree. Choi et al. [15] indicated that privacy fatigue may be the main problem preventing people from following required security measures and practices. Moreover, Coopamootoo et al. [16] suggested that privacy fatigue could also affect subsequent tasks.

Privacy behavior often depends on users' attitudes and emotions [45]. Existing studies in the field of human-computer interaction (e.g., CHI and CSCW) have explored different ways to guide users in different privacy environments [65, 67]. A growing number of empirical studies have begun to emphasize the role of such systemic inconsistencies in privacy decisions, and attempt to mitigate users' perception of fatigue with interventions such as available security policies or interaction design, and to promote users' security decisions [5, 21, 28, 46, 72, 73]

Although privacy fatigue is being studied already, it is far from enough. In particular, we need to study from the perspective of psychology and behavioral economics. Related concepts and scales are only preliminary cognition levels. To fully understand and verify the privacy fatigue of social media users, other antecedent variables and behaviors caused by fatigue are worth further study. Moreover, existing literature does not consider whether different personality characteristics will moderate users' privacy fatigue perception under the same conditions, and a relatively perfect cognitive model or theoretical framework has not been established.

3 METHODOLOGY

We selected Weibo users as experimental participants. Weibo is one of the mainstream social media in China with 566 million users [2]. Secondly, many studies in the field of HCI have been carried out with Weibo as the background [11, 55]. To ensure the content validity of the scale, 15 participants were randomly selected for the pre-experiment. The formal experiment was conducted in July and August 2021, and there were 428 valid participants. The number of questionnaires collected is in line with the quantity required for structural equation modeling and analysis, which can ensure the stability of samples to a large extent and facilitate subsequent structural equation data analysis [64]. The basic information of the sample population is as follows: male users are 45.3% and female users are 54.7%. Among them, 87.4% of the respondents are between 18 and 45 years old. All participants have been using social media for more than five years. They spend an average of 3.7 hours and browse social media more than three times a day on the Internet.

Most of the variables refer to the maturity scale of relevant studies, and are localized adjusted and improved according to the research situation [10, 13, 15, 20, 42]. Especially for the personality prediction scale, we selected the Chinese Big Five Personality Scale (CBF-PI-B) suitable for the local context, which has been proved to accord with the unique culture and environment of China [70].

3.1 Develop hypothesis and Construct S-O-R model

In this chapter, we will specifically describe the hypothesis formulation and the construction of the S-O-R model.

3.1.1 Privacy fatigue. Fatigue has been extensively studied in the fields of medicine, financial analysis, and decision-making. It is manifested as a subjective and unpleasant feeling [8, 49, 52]. The specific phenomenon that maps to cyberspace security behavior is privacy fatigue. Privacy fatigue arises from two major reasons: one is that personal information becomes difficult to manage and how it is used cannot be controlled, and the other is that the protection measures provided by the network platform are too complicated to be understood. These all make users feel tired to cope with the fatigue [29]. Therefore, privacy fatigue has the attributes of "forced acceptance" or "helpless obedience", reflecting the users' boredom emotion with privacy decisions. Only after correctly understanding the causes and consequences that lead to privacy fatigue can further measures be taken to alleviate user privacy fatigue.

3.1.2 Information overload and Perceived privacy risks. Psychologists have been aware that information overload can affect individual performance and decision-making ability [63]. Excessive complexity and strict privacy protection can exacerbate information overload [3, 9]. For example, the existing privacy policies and related regulations are complex and lengthy, resulting in information overload. They become tired and lose control of their privacy decisions.

In addition to the objective environmental attitude based on information overload, personal negative emotions, such as anxiety, depression, and pessimism about privacy decisions, may affect their privacy decisions at the subconscious level [68]. We attempted to analyze the causes and influencing factors of privacy fatigue from users' internal psychological motivation [15, 68]. Social media users experience the mismatch between environmental needs and personal privacy needs, which may make users more prone to privacy fatigue. Therefore, we propose the following assumptions from both the external objective environment and internal subjective factors:

H1: Information overload is positively correlated with privacy fatigue

H2: Perceived privacy risk is positively correlated with privacy fatigue

3.1.3 Disengagement behavior and Willingness to disclose information. Fatigue will have a significant impact on individual behavioral decisions, leading to people's desire to minimize decision-making efforts [19]. Privacy fatigue may increase users' cognitive load, anxiety, and stress levels, which leads to excessive self-disclosure results [50, 58]. Accordingly, fatigued users tend to choose to accept default options for managing privacy and publish information without careful consideration when making self-disclosure [35].

Meanwhile, previous studies indicated that disengagement is one of the key outcomes of fatigue [31, 51, 53]. This means that privacy fatigue may lead users to reduce their frequency of social networking sites, or even abandon them [7, 34]. Valero classified three discontinuous behaviors on Social Networking Services (SNS) as controlling activities, short breaks, and pause behaviors [51].

As a result, privacy fatigue may have a particularly strong effect on disengagement behavior. Fatigued users are more likely to be "hands-off" about the abuse of personal information.

Therefore, based on the previous extended literature, we assume that

H3: Privacy fatigue is positively correlated with excessive disclosure

H4: Privacy fatigue is positively correlated with disengagement

3.1.4 Personality traits. Privacy is considered part of a more general "personality right" and research suggests that personality traits can influence privacy decisions. Particularly, studies have shown personality traits play a key role in forming personal beliefs and motivating personal behaviors [6]. Personality could explain the significant proportion of fatigue variation [18]. Five-factor Model (FFM) is the most classic practice describing five types of personality traits: agreeableness, neuroticism, conscientiousness, extraversion, and openness [47]. Lee conducted a study on Facebook users with different personality traits and found that users' personality traits influenced their social media fatigue [40]. Moreover, personality may be specifically associated with susceptibility, precipitation, and persistence of chronic fatigue syndrome (CFS) [40, 66]. Therefore, we speculate that even with the same external environment and internal perception factors, different personalities may lead to different degrees of privacy fatigue of users. Based on the previous research, we assume that:

H5a: Neuroticism plays a moderating role between information overload and privacy fatigue

H5b: Agreeableness plays a moderating role between information overload and privacy fatigue

H5c: Extraversion plays a moderating role between information overload and privacy fatigue

H5d: Conscientiousness plays a moderating role between information overload and privacy fatigue

H5e: Openness plays a moderating role between information overload and privacy fatigue

H6a: Neuroticism plays a moderating role between perceived privacy risk and privacy fatigue

H6b: Agreeableness plays a moderating role between perceived privacy risk and privacy fatigue

H6c: Extraversion plays a moderating role between perceived privacy risk and privacy fatigue

H6d: Conscientiousness plays a moderating role between perceived privacy risk and privacy fatigue

H6e: Openness plays a moderating role between perceived privacy risk and privacy fatigue

3.1.5 Construct S-O-R Model. The Stimulus Organism Response (S-O-R) framework constructs a mechanism to explain human behavior by analyzing the cognitive state affected by environmental stimuli [44]. It has been applied to online contexts [37], analyzing the reactions of social media users [36] and human-computer interaction [69]. Therefore, we extend the use of the S-O-R framework to the theoretical research on privacy fatigue of Weibo, a specific and popular social media platform in China.

According to the definition of the S-O-R framework [33, 41, 56], we define stimuli as objective environmental factors of information

Table 1: Analysis results of structural equation model (p<0.01; *p<0.05; factor1-factor5: Information overload, Perceived Privacy Risks, Privacy fatigue, Disengagement behavior, Willingness to disclose information)**

X	Y	z(CR)	Standardized regression coefficient	result
H1: factor1 →	factor3	12.531	0.731**	support
H2: factor2 →	factor3	4.816	0.675**	support
H3: factor3 →	factor4	14.249	0.679**	support
H4: factor3 →	factor5	12.22	0.691**	support

Table 2: Structural equation model fitting index

Common index	χ^2/df	GFI	RMSEA	CFI	NFI
Measurement criteria	<3	>0.9	<0.10	>0.9	>0.9
Value	2.793	0.931	0.008	0.915	0.921

overload and personal subjective factors of perceived privacy risk. We define "organism" as the intermediary state of human emotional cognition, privacy fatigue, which explains the impact of stimuli on individual behavioral consequences. "Reaction" is defined as the over-disclosure and disengagement that may occur after users' privacy fatigue. Consequently, based on the S-O-R model, we propose an overall framework to explain how environmental factors and user perception affect user behavior through privacy fatigue. The specific model is shown in Figure 1.

3.2 RESULTS

3.2.1 Reliability analysis and Validity analysis. The scales learn from the literature in a non-Chinese environment and are modified according to our theme. Therefore, we need to focus on whether localization of the scale is feasible. We measure the reliability and validity firstly. The reliability coefficient of each variable is greater than 0.8, so it can also be considered that the reliability is higher and the effect is better [64]. The Kaiser-Meyer-Olkin (KMO) value is 0.912, indicating that the data is suitable for subsequent factor analysis [25, 32]. We further measure the convergence validity and discriminative validity, showing that the measurement model is appropriate [25].

3.2.2 Construct structural equation model (SEM). We test the proposed model using Structural Equation Modeling supported by Partial Least Squares (PLS), which has been widely adopted in the field of information systems [25, 60]. The significance represents the mutual influence between the paths and the path coefficient reflects the strength of the hypotheses. The analysis results of the proposed model are shown in Table 1. Furthermore, the structural equation model fitting index is shown in Table 2, indicating the model index fits well.

The results show that all the hypotheses proposed in the structural equation model are supported. Specifically, as expected, the study confirmed that privacy fatigue plays a vital role in understanding the information behavior of application users. The p-values of all path coefficients in the research model are significant, indicating that the hypotheses have passed the significance test. Information

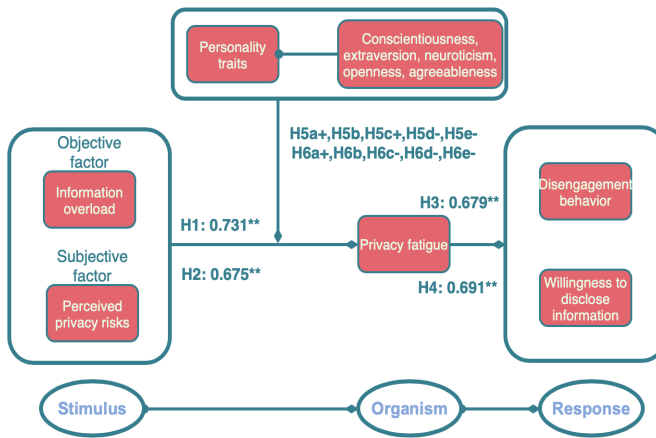


Figure 1: Structural equation model based on S-O-R theory ($p < 0.01$; * $p < 0.05$)**

overload and perceived privacy risks will significantly affect the privacy fatigue perception of social media users, supporting H1 and H2. Privacy fatigue will significantly lead to more willingness to disclose and disengagement, supporting H3 and H4.

More specifically, the impact of information overload on social media users' privacy fatigue is greater than the perceived privacy risk. Namely, compared with the subjective attitude of privacy perceived risk, users' negative fatigue emotion caused by information overload are more likely to cause privacy fatigue decisions. On the other hand, fatigue also has a significant impact on the intention to disclosure and disengagement. The impact of privacy fatigue on over disclosure intention is greater than disengagement. The results also indicate that users with high privacy fatigue tend to make fewer privacy decisions. When users feel irreversibly losing control of their privacy, they will be reluctant to go any further and simply give up protecting their privacy and security. In this case, they are likely to mechanically leak privacy information, leading to a vicious circle. Moreover, users with high privacy fatigue are more likely to "do nothing" about the abuse of personal privacy. Overall, the results demonstrate the important role of privacy fatigue in users' social privacy behavior and validate our proposed related hypotheses.

3.2.3 Moderating effect of personality. As discussed above, we take personality traits as the moderating variable between fatigue antecedents and privacy fatigue, trying to verify whether different personalities have moderating effects under the same perception. Personality score is used as quantitative data. Therefore, we first conduct data centralized processing to make the analysis results more stable and interpretable [17]. The data analysis results of the entire structural equation model based on the S-O-R theory are shown in Figure 1.

The data analysis results are shown in Table 3. In addition to agreeableness, personality traits have a significant moderating effect between perceived privacy fatigue and privacy fatigue. However, the analysis results may be different from the previous research

results. Specifically, neuroticism strengthens the relationship between information overload, perceived privacy risk, and privacy fatigue. Therefore, H5a and H6a are supported representing the moderating effect is positive. This is similar to previous research conclusions. Neuroticism reflects the trend of personal experience of nervous tension, depression, internalization and self-awareness and plays a role in the stressors affecting social media fatigue [43], which will positively promote users' privacy fatigue. Neurotic users are more sensitive to objective environmental factors and personal subjective factors, feeling uneasy because of the blurred boundaries of personal privacy. Their sense of privacy fatigue could be exacerbated by information overload and perceived privacy risks.

Extroversion plays a moderating role in information overload, perceived privacy risk, and privacy fatigue. The specific adjustments show different directions. Specifically, H5c is supported by the results, and its moderating effects are positive. The reason may be that people with high extroversion scores tend to respond positively to stimuli [26]. The result is similar to Shokouyar's finding that extroversion affects users' social media fatigue behavior [57]. However, the information overload caused by the complex privacy policies and interactions provided by social media platforms is overwhelming for extroverts, reflecting users' fatigue over privacy decisions. What is inconsistent is that H6c is supported by the results, and the moderating effects are negative. Other studies suggest that extraversion was negatively correlated with emotional exhaustion [30]. The reason may be that extroverts are more enthusiastic, confident, and focus on pursuing enjoyment values. Therefore, even if extroverts perceive privacy risks, they are more likely to continue to keep in touch with friends through social media, and they are less concerned about whether their privacy is leaked.

Conscientiousness plays a significant role in mediating information overload, perceived privacy risk, and privacy fatigue. Therefore, H5d and H6d are supported by the results, and the moderating effect is negative. Conscientious individuals usually remain cautious and vigilant when predicting the consequences of their decisions in advance [39]. Therefore, when disclosing personal information, they tend to pay more attention to how service providers use and process their personal information. For them, authorizing or releasing private information is prudent and less prone to perceived privacy fatigue.

However, it's worth noting that agreeableness has no mediating effect among information overload, perceived privacy risk, and privacy fatigue. The result is consistent with Lee's research, the agreeableness traits of employees were not associated with work fatigue [48]. Nevertheless, the result is contrary to Swider's research [59]. Agreeableness is also negatively correlated with emotional exhaustion and lack of personal accomplishment. It may be because agreeable people have a strong affinity and are less concerned about the risks of using SNS, so they are not sensitive to the perception of privacy fatigue. Therefore, H5b and H6b were rejected by the result.

Openness also has a remarkable moderating effect among information overload, perceived privacy risk, and privacy fatigue. Therefore, H5e and H6e are supported by the results, and their effects are negative. Similar to Shokouyar's research, openness affects users' social media fatigue behavior [57]. This may be because

Table 3: Moderating effects analysis results **p<0.01 *p<0.05 (factor1: Information overload; factor2: Perceived privacy risks)

Hypothesis		<i>p</i>	LLCI	ULCI	Hypothesis		<i>p</i>	LLCI	ULCI
H5a	factor1	.0000**	0.7971	0.9387	H6a	factor2	.0000**	.6757	.8046
	neuroticism	.0000**	0.0698	0.0967		neuroticism	.0000**	-.0607	-.0886
	factor1*neuroticism	.0033**	0.0062	0.0311		factor2*neuroticism	.0114*	.0033	.0257
H5b	factor1	.0322*	.6719	.8291	H6b	factor2	.0208*	.5918	.7331
	agreeableness	.0212*	-.0823	-.0538		agreeableness	.0283*	-.0755	-.0468
	factor1*agreeableness	.0569	-.0241	-.0002		factor*agreeableness	.5168	-.0149	.0075
H5c	factor1	.0000**	.2070	.4238	H6c	factor2	.0046**	.0422	.2308
	extraversion	.0023**	.1034	.1329		extraversion	.0048**	.1159	.1475
	factor1*extraversion	.0027**	.0058	.0317		factor2*extraversion	.0405*	-.0217	-.0005
H5d	factor1	.0000**	.2665	.4567	H6d	factor2	.0000**	.1429	.3100
	conscientiousness	.0000**	-.1342	-.1074		conscientiousness	.0000**	-.1367	-.1083
	factor1*conscientiousness	.0000**	-.0489	-.0276		factor2*conscientiousness	.0200*	-.0211	-.0018
H5e	factor1	.0000**	.3232	.5029	H6e	factor2	.0000**	.2201	.3746
	openness	.0000**	-.1264	-.1010		openness	.0000**	-.1278	-.1018
	factor1*openness	.0000**	-.0480	-.0280		factor2*openness	.0016**	-.0240	-.0056

people with open personalities are less likely to have rigid conscientiousness or behavior [14]. Open people are sensitive to social concerns but are less likely to give in to pressure. They can treat the privacy decision of social media more rationally. Meanwhile, they adopt an optimistic attitude towards the accompanying privacy interaction and decision, so they will not be prone to perceive privacy fatigue.

Through the above discussion, we can conclude that there are significant differences in the influence of personality traits on privacy fatigue. Correspondingly, the data analysis results are not the same as previous research conclusions and there are certain differences. This may be due to different experimental environments and cultural backgrounds. Therefore, it is not enough to rely on previous studies in other contexts to understand the relationship between social media users' personality characteristics and fatigue. Our work attempts to supplement and expand existing research.

4 DISCUSSION

The research combines several complementary research fields. We establish a relatively complete fatigue cognitive model and theoretical framework, focusing on the perceived antecedents and behavioral consequences. Furthermore, we verify the moderating effect of the users' different personality traits. The study corresponds to Lee et al. [39] called for research to explore individual differences that may cause social media fatigue. As far as we know, the study is the first work to understand privacy fatigue by verifying the moderating role of personality characteristics in the relationship between driving antecedents and privacy fatigue. Although the research is exploratory in nature, the results show that there does seem to be a relationship between the established frameworks. For example, compared with perceived privacy risks, information overload has a greater impact on privacy fatigue. Moreover, part of personality traits has a moderating effect on social media privacy fatigue, indicating that the privacy interaction design of intervention strategies in SNS should conform to the characteristics of different users to maximize its effectiveness and usability.

Further research could expand the scope by exploring other factors to enhance the effectiveness and explanatory capabilities

of the model. Studies showed that there are differences in privacy perception in individualistic and collectivist cultures [71]. Recent work on the contextual online privacy perception model has also shown that nationality has a moderating effect on privacy behavior [12]. Therefore, future studies could be based on other social media platforms or replicate this research in other cultural contexts to improve the generalization of the model and results.

5 CONCLUSION

Although most previous studies have focused on the benefits of social media, it seems essential to investigate the possible privacy dark side. The research provides important theoretical supplementary significance for a better understanding of online privacy behavior. The results support the opinion of Choi et al. [15] that privacy fatigue is an important factor and a new perspective to explain the privacy paradox. We hope that the study will provide a more comprehensive understanding of social media privacy fatigue, including perceived antecedents, behavioral consequences, and different personality traits factors.

Accordingly, we suggest that online service providers should be aware of the potential impact of privacy fatigue on services, and be cautious about factors that can cause significant fatigue to users. It is important to purposefully design and develop interfaces that may bring minimal privacy fatigue to users. Laws such as the general data protection regulations (GDPR) emphasize the usability of privacy notification and selection. Nevertheless, most users do not understand the notice or policy and feel tired, thinking that they are protected by non-existent laws. Hence, the regulations should consider security and privacy decisions from the user's perspective. For policymakers, it is also necessary to strengthen the regulatory responsibilities of government management departments in terms of cyberspace privacy. Of course, fatigue is not the only factor that affects privacy and security compliance, and user behavior will also be affected by other factors. However, fatigue is likely to occur, so it becomes another parameter that must be considered in the struggle to maintain effective privacy protection.

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